WORKED SOLUTION

STEP II 2003, Question 7

Pure • Improper Integrals and Series

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s2-q7

The key idea

An integrand of the form $\ln x \times (\text{power of } x)$ is made for integration by parts: differentiate the logarithm and it disappears. That handles the first integral. The rest of the question turns the exact value $2/(n^2\mathrm{e})$ round: it lets us *write* $1/n^2$ as an integral. Then we move all the lower limits to the same place, which can only make each integral bigger, and add up a geometric series inside the integral.

Solution

The integral

For $a > 1$ and $n > 0$ let

$$I_n(a) = \int_a^\infty \frac{\ln x}{x^{n+1}}\,\mathrm{d}x.$$

Integrate by parts with $u = \ln x$ and $\mathrm{d}v = x^{-n-1}\,\mathrm{d}x$, so $v = -\frac{1}{n}x^{-n}$ (this is where we use $n \neq 0$). Over $[a, X]$,

$$\begin{aligned}\int_a^X \frac{\ln x}{x^{n+1}}\,\mathrm{d}x &= \left[-\frac{\ln x}{nx^n}\right]_a^X + \frac{1}{n}\int_a^X x^{-n-1}\,\mathrm{d}x \\ &= \frac{\ln a}{na^n} - \frac{\ln X}{nX^n} + \frac{1}{n^2}\left(\frac{1}{a^n} - \frac{1}{X^n}\right).\end{aligned}$$

Now let $X \to \infty$. Since $n > 0$, $X^{-n} \to 0$. For the other term, put $Y = X^n$, so that $\ln X = \frac{1}{n}\ln Y$ and

$$\frac{\ln X}{X^n} = \frac{1}{n}\cdot\frac{\ln Y}{Y} \to 0,$$

because $Y \to \infty$ as $X \to \infty$ and we are allowed to assume $(\ln Y)/Y \to 0$. (We can't just quote $(\ln X)/X \to 0$ here: when $0 < n < 1$, X^n grows more slowly than X.) So the integral converges and

$$I_n(a) = \frac{\ln a}{na^n} + \frac{1}{n^2 a^n}.$$

At $a = \mathrm{e}^{1/n}$ we have $a^n = \mathrm{e}$ and $\ln a = 1/n$, so

$$I_n(\mathrm{e}^{1/n}) = \frac{1}{n^2\mathrm{e}} + \frac{1}{n^2\mathrm{e}} = \frac{2}{n^2\mathrm{e}},$$

which is the required result.

Comparing the two integrals

If $1 < a < b$, then

$$I_n(a) - I_n(b) = \int_a^b \frac{\ln x}{x^{n+1}}\,\mathrm{d}x.$$

For $x > 1$ we have $\ln x > 0$ and $x^{n+1} > 0$, so the integrand is strictly positive on $[a, b]$. It is also continuous, and $b > a$, so this integral is strictly positive. Hence $I_n(b) < I_n(a)$: moving the lower limit to the right throws away a piece of positive area.

The inequality

By the first part, for every positive integer n,

$$\frac{1}{n^2} = \frac{\mathrm{e}}{2}\, I_n(\mathrm{e}^{1/n}).$$

Ask yourself The sum has a different lower limit $\mathrm{e}^{1/n}$ in each term, but the right-hand side has a single lower limit $\mathrm{e}^{1/N}$. How do we get from one to the other?

Take $1 \le n \le N$. Then $1/N \le 1/n$, so $1 < \mathrm{e}^{1/N} \le \mathrm{e}^{1/n}$. When $n < N$ the inequality is strict, and the second part (with $a = \mathrm{e}^{1/N}$, $b = \mathrm{e}^{1/n}$) gives

$$I_n(\mathrm{e}^{1/n}) < I_n(\mathrm{e}^{1/N});$$

when $n = N$ the two sides are equal. Adding over $n = 1, \ldots, N$, the sum is strict because $N > 1$ means at least one term (for instance $n = 1$) has $n < N$:

$$\sum_{n=1}^{N} \frac{1}{n^2} = \frac{\mathrm{e}}{2} \sum_{n=1}^{N} I_n(\mathrm{e}^{1/n}) < \frac{\mathrm{e}}{2} \sum_{n=1}^{N} I_n(\mathrm{e}^{1/N}).$$

This is the point where we need $N > 1$. With $N = 1$ the two sides would be equal.

It remains to add up the integrals. A finite sum of convergent integrals over the same range is the integral of the sum, so

$$\sum_{n=1}^{N} I_n(\mathrm{e}^{1/N}) = \int_{\mathrm{e}^{1/N}}^{\infty} \left(\sum_{n=1}^{N} x^{-(n+1)} \right) \ln x\,\mathrm{d}x.$$

The bracket is a geometric series with first term x^{-2}, ratio $x^{-1} \neq 1$ (as $x > 1$) and N terms:

$$\sum_{n=1}^{N} x^{-(n+1)} = \frac{x^{-2}(1 - x^{-N})}{1 - x^{-1}} = \frac{1 - x^{-N}}{x^2 - x},$$

after multiplying top and bottom by x^2. Therefore

$$\sum_{n=1}^{N} \frac{1}{n^2} < \frac{\mathrm{e}}{2} \int_{\mathrm{e}^{1/N}}^{\infty} \left(\frac{1 - x^{-N}}{x^2 - x} \right) \ln x \, \mathrm{d}x,$$

as required.

Checks

- Differentiate the formula for $I_n(a)$ with respect to a: you should get $-\ln a / a^{n+1}$, the negative of the integrand, as the fundamental theorem of calculus demands.
- Numerically, with the substitution $x = \mathrm{e}^u$ the integral in the final part becomes $\int_{1/N}^{\infty} u(1 - \mathrm{e}^{-Nu})/(\mathrm{e}^u - 1) \, \mathrm{d}u$. For $N = 2$ the right-hand side is about 1.4865, against $1 + \frac{1}{4} = 1.25$. For $N = 10$ it is about 2.0579, against 1.5498.
- As $N \to \infty$ the right-hand side tends to $\frac{\mathrm{e}}{2} \int_1^{\infty} \frac{\ln x}{x^2 - x} \, \mathrm{d}x$, and the substitution $x = 1/t$ turns this into $\frac{\mathrm{e}}{2} \int_0^1 \frac{-\ln t}{1-t} \, \mathrm{d}t = \frac{\mathrm{e}}{2} \cdot \frac{\pi^2}{6} \approx 2.236$. This is comfortably above $\pi^2/6 \approx 1.645$, as it must be.

Common slips

- Using $(\ln X)/X \to 0$ directly to kill $(\ln X)/X^n$. That works for $n \geq 1$, but the result is claimed for all $n > 0$.
- Moving the lower limits the wrong way. Since $\mathrm{e}^{1/N}$ is the *smallest* of the $\mathrm{e}^{1/n}$, every integral gets bigger.
- Claiming strict inequality without saying why. The $n = N$ term gives equality, so strictness comes from the terms with $n < N$, which exist because $N > 1$.

Beyond STEP

Writing a number as an integral and then summing under the integral sign is a standard move in analysis. Taken to the limit it gives $\sum 1/n^2 = \int_0^1 \frac{-\ln t}{1-t} \, \mathrm{d}t$, one route to $\pi^2/6$. Letting $N \to \infty$ swaps an infinite sum with an improper integral. That swap needs justification, through the monotone or dominated convergence theorems, which you meet in a first analysis or measure theory course. Abbott's *Understanding Analysis* is a friendly place to see why such interchanges can fail and when they are safe.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.